

Ramesh Alagesan

Senior Data Scientist

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[LinkedIn Profile](#)

Summary

- Accomplished Data Scientist with over 10 years of hands-on experience designing and developing machine learning models, backed by 20 years of technology industry experience.
- Proficient in Machine Learning, Data Analysis, and Data Mining, with deep expertise in managing large-scale structured and unstructured datasets. Proven track record of building predictive models across diverse sectors, including healthcare and life sciences.
- Adept in analyzing and presenting complex data with proven problem solving abilities.
- Extensive experience with Natural Language Processing (NLP), Machine Learning (ML) Algorithms, Classification Models, Generative Pre-Trained Models (GenAI), Large Language Models (LLMs), Retrieval Augmented Generations (RAG) and Prompt Engineering.
- Strong coding skills in Python, with a track record of delivering solutions to production.
- Implemented techniques to optimize model performance to meet specific product goals.
- Skilled in NLP methods for information extraction, topic modeling, sentiment analysis, parsing, and relationship extraction. with a track record of developing, deploying, and maintaining production NLP models using NLTK, Text blob, and spacy
- Skilled in architecting overall frameworks for GenAI products such as search interfaces, bots, summarizers, etc.

Skills

- Machine Learning: LLM, Natural Language Processing, Deep Learning, Classification, Clustering, Sentiment Analysis, Time Series, Regression
- Data Science Tools: Python, R, ChatGPT 3.5/4.0, DBRX, GTE, LangChain, Pinecone, ChormaDB, Milvus, Java, SQL, Hive, Spark ML, AWS Sagemaker, Cloud Platforms (AWS, IBM, Databricks)

Experience

Sr. Data Scientist, Cognizant (Aug 2018 – Jul 2025), Raleigh, NC

Clients: Reynolds,DineBrands, CVS Pharmacy, AbbVie, Bayers, PepsiCo, 3M, Verizon

- Spearheaded multiple GenAI solutions at AI Foundry, including a Personal Assistant using Microsoft Copilot Studio, a ServiceNow Chat Assistant with AWS Q for Business, and a Production Recommendation Engine for the Dines website using Google Retail API.
- Designed and deployed personalized marketing and customer insight tools using AWS Bedrock foundation models, including a marketing email generator and a review analysis solution integrated with BI dashboards.

- Developed a Retrieval-Augmented Generation (RAG) chatbot pipeline using LangChain agents to dynamically retrieve and validate input parameters for 15k reports based on user interaction. Implemented an agent-driven system to manage conditional dependencies between report parameters, ensuring accurate report retrieval by dynamically prompting users for required inputs. Optimized retrieval accuracy through embedding-based search techniques, enhancing report ranking and relevance for user queries.
- Developed a GenAI-driven Sales Promotion Insights conversation agent that combines document, API, and SQL agent functionalities to empower sales personnel with easy access to sales and customer insights via natural language queries. Implemented a robust LLM pipeline that interprets user requests, routes them to client applications, and formats API responses into user-friendly language. Leveraged RAG architecture, OpenAI, Langchain, and vector databases to enhance the bot's performance and response accuracy. This solution simplified the access process, significantly reduced the learning curve for sales teams, and streamlined administrative overhead by 90%, ultimately boosting productivity.
- Developed a predictive model to identify key factors influencing Medicare member Star Ratings, using over 1,000 features from CAHPS, HOS, claims, patient history, and demographic data. Applied advanced feature engineering and addressed class imbalance with techniques like SMOTE and class weighting. The model provided actionable insights to improve member engagement and healthcare quality, helping providers prioritize interventions that lead to higher satisfaction ratings. Tools used include Python, scikit-learn, and SHAP for interpretability.
- Developed a predictive propensity-to-reach model designed to identify the top Medicare members most likely to enroll in preventive care plans from a pool of 3 million members. The model successfully identified the top 20% of members who were most likely to enroll in preventive care plans, achieving a significant improvement in outreach success rates. The personalized outreach scheduling based on the model's time predictions led to a 30% increase in engagement compared to standard outreach methods.
- Developed a machine learning model to predict osteoporosis risk using key factors such as age, gender, BMI, lifestyle habits, medical history, and bone mineral density measurements. The model provided accurate risk scores, helping identify high-risk individuals for early intervention.
- Improved member satisfaction by aligning support calls with predefined client metrics, analyzing call sentiment, and identifying key areas for service enhancement.
- Extracted member sentiments and intents from CAHPS open-text responses to drive actionable insights.
- Developed an image-based recommendation system that extracted product specifications, validated them against an ontology, and accurately identified the products.
- Automated the extraction of project objectives, team roles, costs, tools, sectors, clients, and divisions from sales lead presentations, enabling personalized opportunity recommendations for the sales team.
- Developed an anomaly detection model using AWS Random Cut Forest to identify abnormal state level Medicare financial expense items from 300+ expenses. Created a financial forecast model using multiple machine learning techniques to predict the financial budget amount for each state and line item.
- Developed and implemented CMS budget forecasting solution in AWS Sagemaker for data preprocessing, feature extraction, API calls, and orchestration, Spark, AutoML, model selection, execution plans, API endpoints, and document databases for end-to-end solutions.

Data Scientist, IBM (Jun 2014 - Jul 2018), Research Triangle Park, NC

- Identified supply chain use cases and optimized business value using AI.
- Led the conceptualization, architecture, and implementation of IBM Risk Insights, an NLP-based solution for identifying supply chain risks from news articles, weather events, and social media.
- Designed and implemented a sophisticated resource allocation model for IBM's Global Business Services (GBS) team, aimed at identifying the optimal method for resource allocation. Leveraged advanced linear programming techniques to formulate and refine the model, ensuring efficiency and precision in resource distribution

- Developed a search engine/recommender system that improved product quality and client satisfaction by recommending fixes for reported defects from machine logs.
- Built a cognitive solution to convert customer invoices from image to text and extract invoice information for ERP applications.
- Developed predictive models for quote and pricing solutions using advanced analytics and AI.
- Tools/ML Methods: Python, R, Java, IBM SPSS, Watson Studio, IBM Watson AI services, NLP, Spark ML, Hive, Sentiment Analysis.

Oracle PeopleSoft CRM Consultant: Accenture, Deloitte, Rexnord (2001 - 2012)

- Transformed Helpdesk business process into CRM application. Led go-live process for a highly visible government project.
- Implemented an integrated business workflow solution for a very large enterprise merger (Wells Fargo and Wachovia) through technology solutions.
- Automated technician dispatch process which improved resource utilization by up to 40%.
- Reduced call center average response time by 50%, and mainframe utilization by improving software performance.

Software Developer, Arkin Systems: (1998 -2001)

- Developed software application to automate healthcare survey OCR data preparation.
- Developed a school management product to efficiently handle teacher management, classroom scheduling, and resource allocation.

Certificates

- AWS ML Specialty Certification
- AWS Data Analytics Specialty Certification
- AWS Cloud Foundations Certification
- IBM Watson Discovery Certification

Education

University of Tennessee – Knoxville, TN | Fall 2012 - Spring 2014

- Master of Science in Business Analytics
- MBA in Supply Chain Management

National Institute of technology, Trichy, India,1998

- Masters of Science in Operations Search and Computer Applications

Bharathidasan University, Trichy, India, 1995

- Bachelor of Science in Mathematics